

Biochemical Guardrails — Q-Twin

Draft prepared from the 45-chip raw dataset (14/15/20/21 Mar 2026). This is a first pass built directly off the computed numbers — Eva should read through, correct anything that conflicts with lab notes, and own the final judgment calls (flagged below).

1. Concentration bounds

Valid range: 0.5–10 μM . Concentrations outside this range are not physically meaningful for this assay and should be excluded from training data for the AI models in Weeks 2+.

DRAFT — needs Eva's review: this range is taken directly from the Week 1 guide (Task 5, item 13). The raw dataset also includes 20 μM , 40 μM and NC chips, used here only to characterize the crowding inversion (Section 2) and negative-control behavior (Section 4) — confirm this is the intended scope, not a contradiction.

2. The 10 μM crowding inversion

Above roughly 10 μM (probe concentration during the probe-deposition stage), Δf stops getting more negative and starts trending positive — surface crowding limits further mass loading, and non-specific/multilayer effects can push the frequency back up instead of down.

Reference number from the published paper/summary: -62.96 Hz at 10 μM , room temperature.

Concentration	Chips (n)	Mean probe Δf (Hz)	Range (Hz)	Note
0 μM	3	-0.72	-1.46 to +0.14	baseline, no probe
5 μM	3	+1.15	-0.81 to +4.39	
10 μM	3	+3.44	+1.94 to +5.19	above crowding onset
20 μM	3	+1.11	-1.12 to +5.12	
40 μM	3	+2.91	-1.48 to +6.98	

Table computed from the 14 Mar concentration-sweep chips (probe-stage Δf , averaged per replicate then per chip). Sample size is small ($n=3$ per concentration) and noisy — the trend is directionally consistent with crowding (Δf becomes less negative / goes positive above ~ 5 –10 μM) but does not cleanly reproduce the -62.96 Hz reference point from the published paper. That number likely comes from a cleaner/aggregated measurement elsewhere in the original dataset.

DRAFT — needs Eva's review: reconcile the -62.96 Hz reference against this table with the teacher — same open question as Task 2 in the Week 1 guide (the 70.18% vs 70.72% mismatch). Also confirm whether 'crowding onset' should be pinned at 10 μM or whether 5 μM already shows early signs (mean here is only weakly negative).

3. Chip failure logic

SUCCESS if probe-stage $\Delta f < 0$ Hz. FAILURE otherwise ($\Delta f \geq 0$, or missing probe data).

Across the 45-chip dataset: 20 SUCCESS, 25 FAILURE. Mean probe Δf for SUCCESS chips is -1.47 Hz (range -5.09 to -0.13); mean for FAILURE chips is $+2.91$ Hz (range $+0.08$ to $+7.81$). The two groups are well separated around zero — the largest gap in the sorted Δf values near the boundary falls between $+0.14$ Hz and $+0.86$ Hz, just above the threshold, supporting 0 Hz as a clean, physically-motivated cutoff (negative Δf = added mass = binding).

DRAFT — needs Eva's review: confirm 0 Hz (not some small negative buffer, e.g. -0.5 Hz, to account for instrument noise) is the right cutoff — a few chips sit close to zero (No.3: $+0.14$, No.9: -0.13 , No.29 NC: $+0.08$) where measurement noise could flip the call.

4. Real negative-control behavior

Chips No.7 and No.29 (21 Mar) are the only genuine 'no target present' runs. Plots: figures/NC_21Mar_No_7.png and figures/NC_21Mar_No_29.png.

Draft description: both NC chips show flat, low-amplitude, monotonic upward drift over their full run (roughly +10–15 Hz over 150–220s) with no sharp steps or large excursions — this looks like slow thermal/baseline drift, not a binding event. By contrast, a genuine hybridization curve (e.g. chip No.2, a SUCCESS chip from the same 21 Mar session) shows a fast initial drop of tens of Hz within the first ~20–30s and sharp step-like jumps later in the run, both absent from the NC curves. Chip No.29's probe-stage replicates are noisier than No.7's, with small dips instead of a smooth drift, but neither shows the large negative excursion characteristic of a SUCCESS chip.

DRAFT — needs Eva's review: this paragraph is Evin's read of the plots, standing in for Eva's Task 4 write-up. Eva should look at the actual figures (not just this description) and confirm/adjust — in particular the 'this is thermal drift, not binding' interpretation is an inference, not a lab-verified fact.

5. NEW — The Novel Kinetic Biomarker

(a) Time window

30 seconds, measured from the start of the probe-stage curve. Computed as the slope of a linear least-squares fit of Resonance_Frequency vs Relative_time over Relative_time $\leq$ 30s.

(b) Why this window avoids the crowding problem

The sampling interval is ~0.6s, so a 30s window still gives ~45-50 points — enough for a stable slope estimate. A 60s window was considered but rejected: at the higher probe concentrations in this dataset (20-40 μ M), the crowding-driven flattening/reversal described in Section 2 can already be influencing the curve well before 60s, which would contaminate the 'early kinetics' signal with the same surface-saturation effect the biomarker is meant to isolate from. 30s stays safely inside the initial near-linear binding regime for all concentrations tested.

(c) Expected behavior

SUCCESS chips: mean binding rate $\approx$ -0.0149 Hz/s (std 0.032) over the 30s window — i.e., frequency dropping right from the start.

FAILURE chips: mean binding rate $\approx$ +0.0187 Hz/s (std 0.030) — flat or drifting upward from the start, consistent with no early binding.

Across concentrations: expect the magnitude of the early binding rate to increase with concentration up to the ~10 μ M crowding onset (more target/probe available to bind faster), then plateau or shrink above that as surface crowding slows the initial adsorption kinetics too, not just the final Δf .

DRAFT — needs Eva's review: the concentration-dependence claim in the last sentence is a hypothesis based on the crowding model in Section 2, not yet directly verified against this dataset's biomarker column (qtwin/data/chip_summary.csv, binding_rate_probe_dfdt_30s) broken out by concentration group — worth a quick check before this goes in the proposal.

Known limitation: chip 21Mar_No.29, a true negative control, has the second-fastest early binding rate in the whole dataset (-0.071 Hz/s) despite a near-zero full-run Δf (correctly FAILURE). This comes from one noisy replicate declining smoothly for 30s while the other replicate is flat with a step artifact — i.e. a single-replicate false positive on the early-window biomarker alone.

DRAFT — needs Eva's review: should Week 2's models require agreement across replicates before trusting the 30s biomarker, rather than the simple per-chip average used here? See clarifying_questions.md item 7.

Sources

- Computed values: `qtwin/data/chip_summary.csv` (45 chips, all 4 sessions).
- Cross-check against Eva's `chip_index.csv`: `qtwin/data/cross_check_report.csv` (0 mismatches).
- Plots: `qtwin/figures/` (`NC_21Mar_No_7.png`, `NC_21Mar_No_29.png`, and grouped condition plots).